Practice questions on Interference

1. What do you mean by interference due to division of amplitude?
2. What is thin film in optics?
3. Using mathematical calculations show that only first two rays interfere in thin film for reflected system?
4. Derive an expression for path difference between rays reflected in thin film of uniform thickness.
5. Obtain condition for maxima and minima for light reflected from thin uniform film.
6. Derive an expression for path difference between rays transmitted in thin film of uniform thickness
7. Obtain condition for maxima and minima for light transmitted from thin uniform film.
8. With the help of mathematical equations, show that the colours which are present in the reflected system are absent in the transmitted system for interference in thin film.
9. The colours which are present in the reflected system are absent in the transmitted system for interference in thin film. Explain.
10. Explain formation of colours in thin film.
11. What is wedge shaped film?
12. Explain (with mathematical formulae), why we get interference fringes as straight, parallel, and equally spaced lines.
13. What are Newton's Rings
14. Explain (with mathematical formulae) why dark and bright concentric circles are formed in Newton's Ring's experiment.